MemoryCircle

P2: Engineering Baseline

Team FrameFour | Prepared 27 October 2026

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Repository: github.com/osakande-lab/memorycircle
Review branch: p2-engineering-baseline

Current evidence: runnable local increment; 16/16 automated local tests pass; production build passes. Supabase integration and hosting are prepared but unverified. This report records the current baseline, not completion of the full P1 plan.

1. Implementation and connection to P1

MemoryCircle is a private photo-sharing web app for small groups. P1 selected US-01, US-02 and US-06 (11 points) for Sprint 1, plus a separate rendering spike, SP-01. The implemented foundation follows that selection.

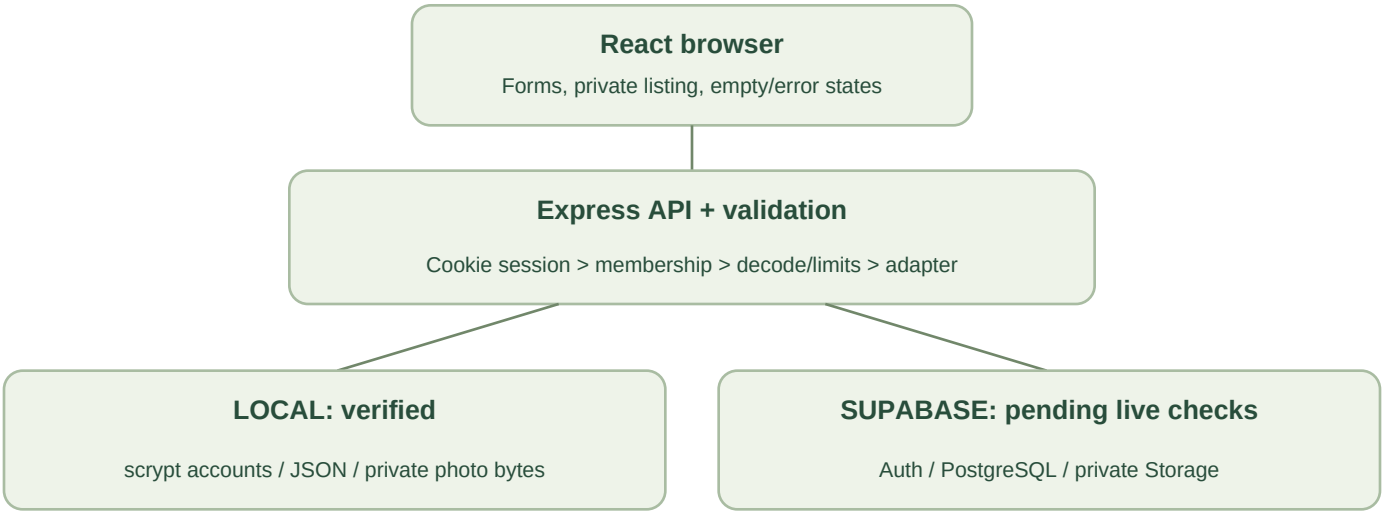
P1 story	Implemented now	Acceptance evidence / gap
US-01 / FR-01	Registration, login, logout, protected routes; revocable eight-hour app sessions.	Local credentials and saved-cookie logout tests pass. Supabase email confirmation and live sessions pending.
US-02 / FR-02	Named daily/weekly private group, persisted owner membership, private access.	Local owner/validation/outsider and persistence tests pass. Supabase transaction not yet executed.
US-06 / FR-06	JPEG/PNG upload, captions, decoded-image and monthly caps, protected metadata/media.	Local happy/negative/concurrency/failure tests pass. Live Storage/RPC acceptance pending.

A minimal selected-month photo list makes accepted uploads visible. It includes an empty state and protected images; it is supporting progress on US-08, not a completed shared-group timeline. Invitations, membership removal, photo editing and recap jobs are not implemented.

Scope adjustment

No Supabase project or intended hosting service exists yet. A clearly labelled local adapter makes the workflow executable without external accounts, while preserving the planned React/Express/Supabase direction. Full Supabase acceptance, deployment and teammate review remain required before the selected stories meet P1's Definition of Done.

2. Initial architectural design



Separate feasibility path: synthetic images + FFmpeg > MP4 (no scheduler/jobs).

Component	Responsibility and repository artifact
React client	client/main.jsx: account/group/upload forms, selected month, feedback. client/style.css: responsive layout and visible focus.
API and validation	server/app.js: auth/private routes, HttpOnly/SameSite cookie, same-origin writes, errors. validation.js: size, caption, actual decoding/re-encoding.
Local adapter	server/local.js: single-process data persistence, scrypt password hashes, opaque sessions, synchronous cap check/save.
Supabase adapter/schema	server/supabase.js and supabase/migrations/001_baseline.sql: backend-only keys, membership checks, private objects, atomic group creation and locked photo publication.

Main engineering decisions

Privacy: private metadata and images pass through authenticated membership-checking routes; media responses use no-store. No service-role key or public bucket URL enters the frontend. Opaque app sessions permit immediate application logout instead of trusting an unexpired provider JWT.

Upload consistency: decode and validate source bytes before saving. Local mode checks/persists synchronously in one process. Supabase writes the object first, then uses a group-row lock to recheck membership and the monthly cap before publication. Failed publication compensates by removing the object; failed cleanup is queued.

Tradeoff: local JSON is demonstration infrastructure and cannot be clustered. Supabase SQL, object compensation and cleanup retry need live verification. The FFmpeg spike remains separate from application recap features.

3. Testing plan and initial results

Method: Node test runner + Supertest exercise the actual Express routes, local adapter and Sharp decoder. Fixtures use synthetic images and isolated data. Windows 10.0.26200; Node v24.22.1. Captured output: tests/latest-results.tap. Commands: **pnpm test** and **pnpm build**.

Area / criteria	Meaningful verification	Result
US-01 AC1-3	Register/login; invalid credentials; signed-out private requests; copied pre-logout cookie rejected; cookie flags.	Pass locally
US-02 AC1-3	Owner ID/schedule persist; invalid input creates no group; outsider denied; restart preserves data.	Pass locally
US-06 AC1-2	Decoded JPEG/PNG, 200-char captions, protected media; corrupt/GIF/missing/>10 MiB/>20 MP/201-char/31st rejected.	Pass locally
US-06 AC3	At 29 photos, two simultaneous uploads return 201 and 409; exactly 30 remain. Real failed file save leaves no completed record.	Pass locally
Other boundaries	UTC month/group cap isolation, cross-site writes denied, internal errors redacted, media no-store.	Pass locally

Total: 16 tests passed, 0 failed, 0 skipped. Build passed locally and on a clean GitHub Linux runner (run 40556656826). These results do not validate the live Supabase adapter, distributed concurrency, hosted deployment, former-member lifecycle or recap privacy.

Example requirement-to-verification chain

US-02 AC3: another user without membership must be denied access to a new private group. **Implementation:** session identity comes from the server credential; group/list/upload/media routes call membership checks. **Test:** an independently registered second account receives 403 for the group, photo list, upload and direct image request; its group list is empty. Signed-out requests receive 401.

Browser and NFR checks

In-app Chromium walkthrough verified registration, sign-in and group creation. Dashboard checks at 390 px and 1366 px showed no horizontal overflow. Chrome/Firefox versions, full keyboard-only flow and photo-loaded UI checks remain pending. NFR-01 load testing, NFR-04 job recovery and NFR-07 cleanup timing are not complete.

SP-01: measured local feasibility

Thirty synthetic 1280 x 720 JPEGs and a generated tone produced a 60-second H.264/AAC MP4. FFmpeg successfully decoded the output and metadata checks passed. Latest elapsed render time: **9.79 seconds**; an earlier run took 11.73 seconds. Host: Intel Core i5-8350U, 8 logical CPUs, approximately 15 GiB memory. Evidence: docs/spike-result.json and docs/ffmpeg.log.

This supports local feasibility only. Docker, intended host, peak memory and the five-run target-host benchmark remain pending. SP-01 is partial; it does not complete automatic recap generation.

4. Updated product backlog

P1 identifiers, relative order, priorities and estimates are retained: **47 points across 12 stories**. No story is Done because P1 requires acceptance/NFR verification and teammate review. In Progress includes local implementation awaiting integration or review.

ID / story	Priority	SP	Status	Current position
US-01 Account access	Blocker	3	In Progress	Local tested; live Supabase and review pending.
US-02 Create private group	Blocker	3	In Progress	Local tested; SQL supplied; live DB/review pending.
US-03 Invite/join	Critical	5	To Do	Expiry/revocation and eight-member cap pending.
US-06 Upload photos	Critical	5	In Progress	Local tested; live storage/concurrency pending.
US-08 Monthly timeline	Critical	3	In Progress	Supporting month list; full shared-member checks pending.
US-04 Leave/remove	Critical	5	To Do	Revocation, jobs and cleanup lifecycle pending.
US-07 Manage own photos	Critical	3	To Do	Edit/delete and recap invalidation pending.
US-10 Automatic recap	Critical	8	To Do	Synthetic spike only; scheduler/pipeline pending.
US-11 Manual recap/status	Critical	5	To Do	Jobs, status, retry and deduplication pending.
US-12 Private recap delivery	Critical	3	To Do	Player/download and ownership checks pending.
US-09 Select music	Major	2	To Do	Licensed catalogue/default/preferences pending.
US-05 Posting schedule	Major	2	To Do	Creation only; updates/contribution status pending.

Significant changes and rationale

The selected Sprint 1 stories now have executable local code and test evidence. A local adapter was added because external services have not been provisioned. A small month-listing view supports upload verification, so US-08 is In Progress without expanding its definition of completion. US-05 remains To Do because saving the initial schedule does not satisfy its update/status criteria.

SP-01 remains separate from the story-point total and only partially verified. Prioritize live Supabase acceptance and teammate review before additional features. Retain original estimates until the team reviews actual integration effort; no velocity or team-hours claim is made.

5. Iteration review and evaluation readiness

Planned versus completed

P1 planned a two-week Sprint 1 with US-01/02/06 (11 points), an eight-team-hour maximum rendering spike, and integration/review. The goal included a deployed private-group/photo foundation. Verified now: local integrated workflow, 16 tests, production build, browser account/group walkthrough, documented architecture, Supabase code/schema and a reproducible local video spike. Deployment and the full Definition of Done remain incomplete.

Difficulties actually observed

Package installation initially attempted a store outside the workspace; a project-local store and hoisted layout resolved it. Persistence tests initially failed in the managed temporary directory; moving isolated fixtures into the workspace allowed real save/restart/failure checks. No Supabase project or hosting choice was available, preventing live integration and target-host evidence. These are observed preparation issues, not reconstructed team collaboration claims.

Responsibilities and contribution evidence

Member	P1 planned responsibility; completion/review not yet verified
Oluwatolayomi Akande	Auth/session/protected routes/sign-in UI; form integration and coordination.
Daniel Seriki	Group/membership schema and creation; authorization/negative-test review.
Nimo Usman	Private photo storage, validation, cap consistency; pair on FFmpeg spike.
Franklin Uba	Upload UI/error states/acceptance tests; lead FFmpeg feasibility.

These are P1 assignments, not proof of completed individual work. This baseline was prepared with AI assistance at the user's request; the user reports instructor permission. Each teammate must independently run, review and explain the work and record actual contributions. No peer reviews, team-hours or cross-member commits have been claimed.

Next iteration adjustments

1. Provision Supabase; run email/session, table/function/bucket denial, storage-failure and independent-process concurrency checks.
2. Complete teammate setup/review and resolve findings before marking stories Done.
3. Select a Docker-capable host, measure runtime/peak memory and five runs; document the proceed/change-host decision.
4. Then implement invitations/shared membership and full timeline, extending privacy checks to later cleanup/recap work.

Repository and TA demonstration

README includes run/test/environment instructions; docs/demo.md gives a repeatable walkthrough. Show the actual local workflow, architecture, automated results and US-02 traceability, followed by backlog and honest development history. The draft branch provides reviewable changes; hosted CI installation/tests/build pass (run 40556656826). Team contributions/reviews and TA access require verification. Do not substitute commit count for evidence. Submit one PDF through D2L after updating any newly completed checks.

Supporting artifacts: docs/architecture.md, docs/testing.md, docs/backlog.md, docs/iteration-review.md, tests/latest-results.tap, docs/spike-result.json. P1 source: team's 4 October 2026 requirements/project plan.